COLLEGE OF LAKE COUNTY

Department of Engineering

EGR 260
Introduction to Circuit Analysis

Lab #1

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1 Objective

The objective of this experiment is to verify Ohm's Law by analyzing the relationship between voltage, current, and resistance in a simple electrical circuit. Experimental results are compared with theoretical and simulated values to evaluate measurement accuracy and sources of error.

2 Equipment

- DC power supply
- Digital multimeter
- Resistors (specified values)
- Breadboard
- Connecting wires

3 Experimental Procedure

3.1 Circuit Schematic

Figure 1 shows the schematic of the circuit used in this experiment.

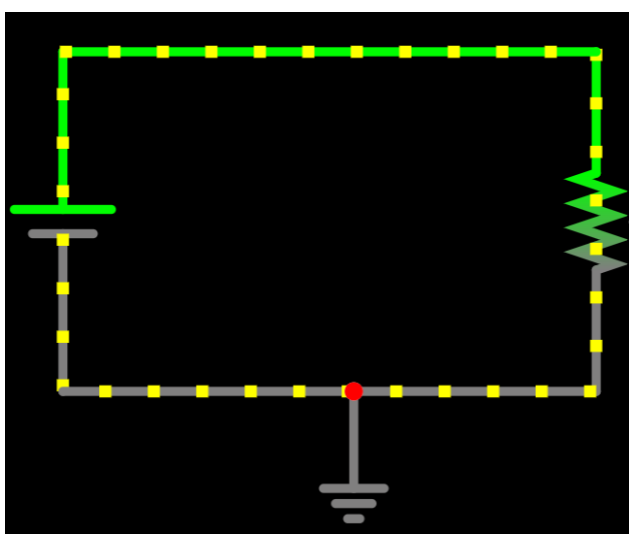

Figure 1: Schematic of simple resistive circuit

3.2 Procedure

1. The circuit was constructed according to the schematic shown in Figure 1.
2. A DC voltage was applied using the power supply.
3. Voltage across the resistor and current through the circuit were measured using a multimeter.
4. Measurements were repeated for multiple voltage levels.

4 Analysis

4.1 Sample Ohm's Law Calculation

Ohm's Law is given by:

$$V = IR \tag{1}$$

Given:

$$V = 1\,\mathrm{V}, \quad R = 1000\,\Omega$$

The theoretical current is:

$$I = \frac{V}{R} = \frac{1}{1000} = 0.001\,\mathrm{A} \tag{2}$$

5 Data

5.1 Measurement of resistor

- R1 = 0.988kΩ normally 1kΩ
- R2 = 2.155kΩ normally 2kΩ
- R3 = 2.856kΩ normally 3kΩ
- R4 = 9.840kΩ normally 10kΩ

5.2 Simulation and measurement values

Voltage (V)	Current (A) in the R1 circuit simulated	Current (A) in the R1 circuit measured
0	0	0
1	0.001	1.002×10^{-3}
2	0.002	2.034×10^{-3}
3	0.003	3.002×10^{-3}
4	0.004	4.075×10^{-3}
5	0.005	5.042×10^{-3}
6	0.006	6.050×10^{-3}
7	0.007	7.066×10^{-3}
8	0.008	8.091×10^{-3}
9	0.009	9.161×10^{-3}
10	0.01	10.129×10^{-3}

Table 1: Voltage and current values for simple 1kΩ resistor circuit

Voltage (V)	Current (A) in the R2 circuit simulated	Current (A) in the R2 circuit measured	Current (A) in the R3 circuit simulated	Current (A) in the R3 circuit measured	Current (A) in the R4 circuit simulated	Current (A) in the R4 circuit measured
0	0	0	0	0	0	0
2	0.001	0.954×10^{-3}	0.667×10^{-3}	0.668×10^{-3}	0.2×10^{-3}	0.208×10^{-3}
4	0.002	1.854×10^{-3}	1.333×10^{-3}	1.348×10^{-3}	0.4×10^{-3}	0.414×10^{-3}
6	0.003	2.819×10^{-3}	2.000×10^{-3}	2.052×10^{-3}	0.6×10^{-3}	0.612×10^{-3}
8	0.004	3.737×10^{-3}	2.667×10^{-3}	2.72×10^{-3}	0.8×10^{-3}	0.822×10^{-3}
10	0.005	4.687×10^{-3}	3.333×10^{-3}	3.41×10^{-3}	1.000×10^{-3}	1.030×10^{-3}

Table 2: Voltage and current values for simple 2kΩ, 3kΩ, 10kΩ resistor circuit

6 Graphs

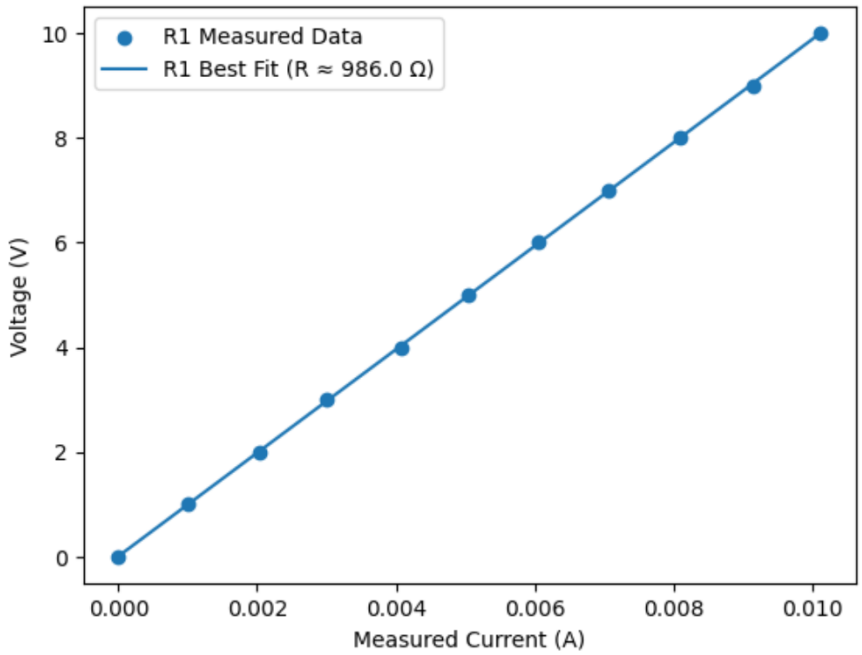

Figure 2: Voltage vs. measured current graph for R1

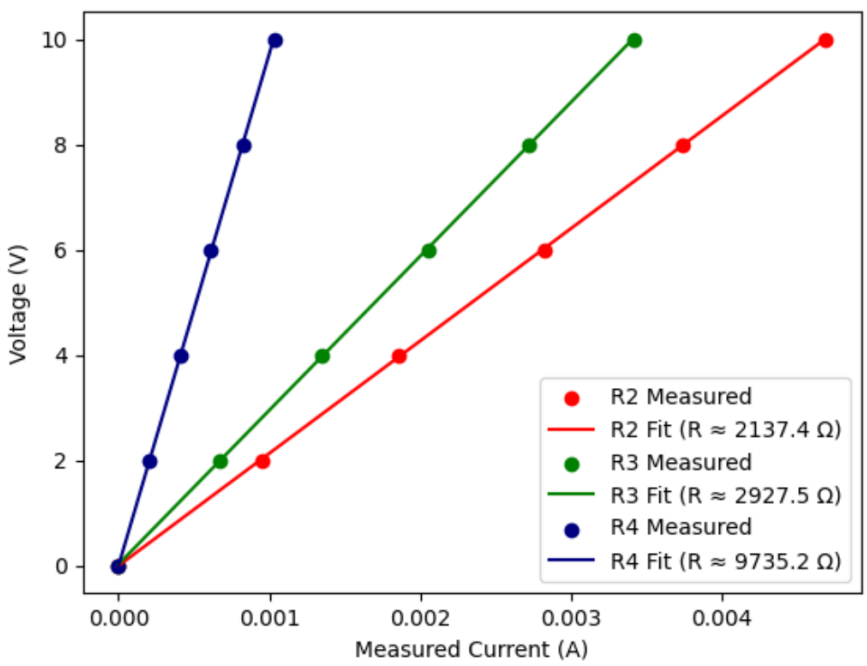

Figure 3: Voltage vs. measured current graph for R2, R3, and R4

7 Discussion and Conclusions

7.1 Relationship

The experimental results in figure 2 and figure 3 demonstrate a linear relationship between voltage and current, confirming that current is linearly proportional to voltage and inversely proportional to resistance. The slope of the voltage-measured current graph represents the values of resistance.

7.2 Experimental and Simulated Results

The measured values were close to simulated results, with minor discrepancies.
Possible sources of error include:

- The difference in initial values of resistors
- Internal resistance of the multimeter
- Contact resistance from wiring and breadboard connections

7.3 Conclusion

This experiment successfully verified Ohm's Law through experimental measurement. The results closely matched theoretical predictions within small discrepancies.